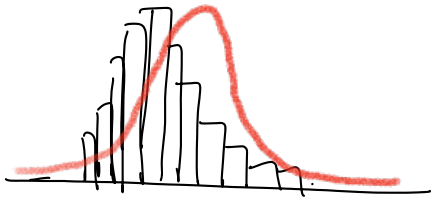
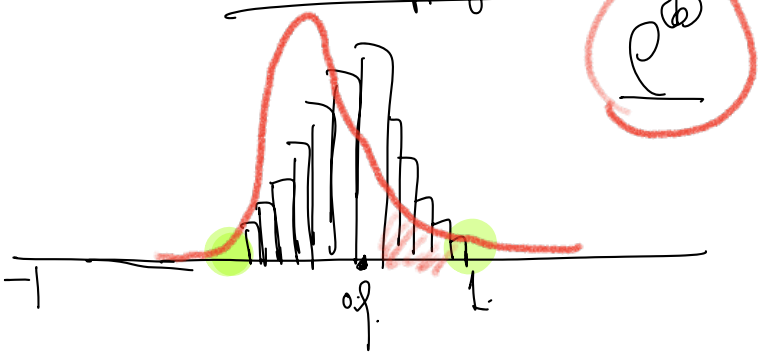
linear regression CI.: $\hat{\beta}_1$

Bootstrapping:



CI: of correlation coefficients:

Bootstrapping:



why skewed?

$\rho_{true} = 0.9$

$-1 \leq \rho \leq 1$ $\quad \rho^{(b)} \leq 1$

$\tilde{x} \sim N(0, \sigma^2)$

$\tilde{\varepsilon} \sim N(0, 1)$

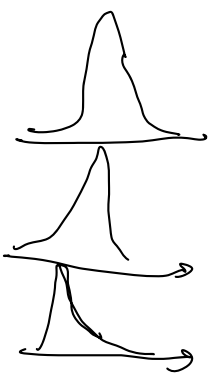
Gen $\tilde{y} = \rho_{true}\tilde{x} + \tilde{\varepsilon}$

Y. X.

0

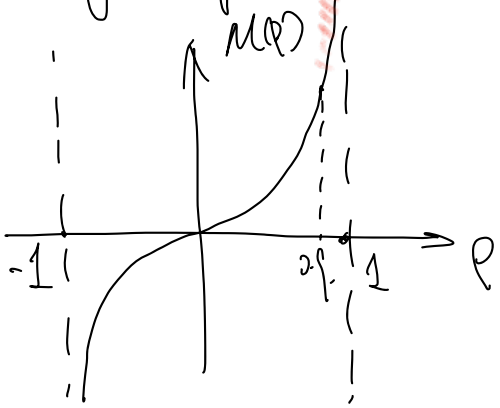
1

-1

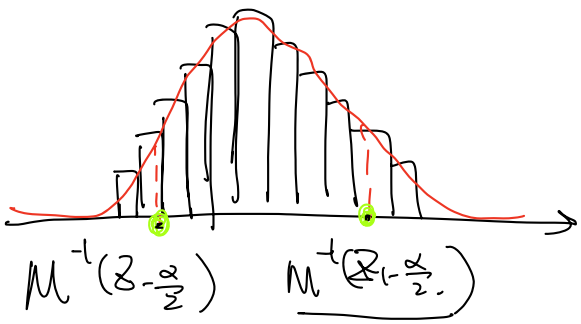


Fisher's transformation:

$M(\rho) = \log \frac{1+\rho}{1-\rho}$



$M(\rho^{(b)})$



$M^{-1}(z_{\frac{\alpha}{2}})$ $\quad M^{-1}(z_{1-\frac{\alpha}{2}})$

① Bootstrap: $\{\hat{\rho}^{(b)}\}_{b=1}^{B}$

② $\rho^{*(b)} = M(\hat{\rho}^{(b)})$ → Fisher. transformation.

③ Gaussian CI on $\{\rho^{*(b)}\}_{b=1}^{B}$ → (L^*, R^*)

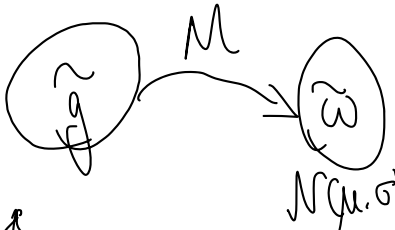
④ Inverse Fisher's trans. $(M^{-1}(L^*), M^{-1}(R^*))$.

Why the new CI works?

Monotone transformation:

Suppose there exists a monotone transformation $M: \mathbb{R} \to \mathbb{R}$.

$\tilde{w} = M(\tilde{g})$ s.t. $\tilde{w} = N(\mu_{\tilde{w}}, \sigma^2)$.



let $w_{\frac{\alpha}{2}} = M(g_{\frac{\alpha}{2}})$. $w_{1-\frac{\alpha}{2}} = M(g_{1-\frac{\alpha}{2}})$.

$P(g_{\frac{\alpha}{2}} \le \tilde{g} \le g_{1-\frac{\alpha}{2}}) = 1-\alpha$?

$= P(M(g_{\frac{\alpha}{2}}) \le M(\tilde{g}) \le M(g_{1-\frac{\alpha}{2}}))$

$= P(w_{\frac{\alpha}{2}} \le \tilde{w} \le w_{1-\frac{\alpha}{2}})$.

$= 1-\alpha$.

$(M^{-1}(w_{\frac{\alpha}{2}}), M^{-1}(w_{1-\frac{\alpha}{2}}))$ ← $(w_{\frac{\alpha}{2}}, w_{1-\frac{\alpha}{2}})$

$P(w_{\frac{\alpha}{2}} \le \tilde{w} \le w_{1-\frac{\alpha}{2}}) = 1-\alpha$

$1-\alpha$.